

Combination of Facial Recognition Models and Explainable AI: A comparison of 2 Deep Fake Detection Algorithms

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Abstract— This project is intended to analyze the sensitivity of a deep neural network (DNN) for facial recognition to fake input images. We began by modifying a transfer-learned DNN that performs facial recognition using weights from a pre-trained ResNet18 model. ResNet18, a specific 18-layer Convolutional Neural Network (CNN) — a subset of DNN — acts as the foundational feature extractor in modern deepfake detection systems. In this context, it functions as a "CNN fake image detector" by identifying subtle, high-frequency texture anomalies, forgery artifacts, and pixel inconsistencies, often achieving high accuracy rates. Torchvision is the most common source, where ResNet18 is available with pre-trained weights. Next, we are going to use Efficientnet-B0, which is a highly efficient convolutional neural network (CNN) that balances speed and accuracy using compound scaling. As the baseline in the EfficientNet family, it serves as a powerful general-purpose backbone for image classification and feature extraction tasks. EfficientNet-B0 models are widely available with pre-trained weights. They are commonly used for transfer-learning. It loads with pytorch. The model training is very slow on CPU even just for 1 epoch. However, it is very fast using a T4 GPU.

I. INTRODUCTION

Deepfakes, primarily generated using Deep Neural Networks such as Generative Adversarial Networks (GANs) and Convolutional Neural Networks (CNNs), represent a significant advancement in synthetic media. As of early 2026, these tools

are capable of creating highly realistic images, videos, and audio, posing substantial risks of forgery and fraudulent activities. While sometimes used for legitimate purposes like film restoration, the potential for misuse in generating deceptive content makes advanced detection critical. Deepfakes, while visually convincing, leave digital signatures—"convolutional traces"—left by the GAN-based generation process. ResNet18 excels at detecting these subtle convolutional operator artifacts.

- ResNet18 utilizes residual learning (skip connections) to train deeper networks without vanishing gradient issues, making it more efficient and accurate than standard, deep neural networks (DNNs). It is used as a binary classifier to distinguish between "Real" and "Fake" (e.g., GAN-generated) faces. The network analyzes features like eye regions or skin texture to identify inconsistencies created by Generative Adversarial Networks (GANs). Rather than training from scratch, ResNet18 is typically pre-trained on ImageNet and fine-tuned for deepfake detection, allowing it to detect fake images with high accuracy on datasets like FaceForensics — this is transfer learning.

The ResNet18 model acts as a feature extractor to identify key anomalies in facial expressions, skin texture, and eye regions, which are then used by the classifier at final layers determining "Real" or "Fake".

We are combining Grad-CAM model to explain the inconsistencies that ResNet18 captures. It works well with Rsnet18 without needing to change the model or retrain it. It generates heatmaps that allow for intuitive visual verification that influences a convolutional neural networks' classification decision.

Deepfake identifying algorithms are better with Explainable AI (XAI) because they transform "black Box" models into transparent tools, allowing for higher accuracy, improved debugging, and increased trust. By highlighting specific, forged facial features (such as pixel irregularities) rather than just giving a probability score, XAI helps identify model biases, fix them, and provide forensic evidence. A "heatmap" or visual justification, which is essential for audit trails, expert review, and establishing trustworthiness in sensitive situations.

Let's look at a different model for facial recognition — EfficientNet-B0 is said to outperform Resnet18 for detecting fake images (deepfake). It could be slower than ResNet18 — but superior in terms of accuracy and precision. Rsnet family of models, including Rsnet18 struggles to capture fine-grained details necessary for high-quality forgery detection compared to EfficientNet's compound scaling approach. As a benchmark, EfficientNet-B0 often exceeds 90 - 94% accuracy, while Rsnet can struggle around 84%. EfficientNet maintains a better balance between accuracy and parameter count, often being faster at inference while providing better accuracy than Rsnet18 or even larger such models. Rsnet can suffer from high validation loss fluctuations. This makes it less effective at generalizing forged datasets compared to EfficientNet-B0.

EfficientNet is designed with neural architecture search and contains 237-layers CNN. We are going to explain the output of this model, also, with Grad-CAM.

The aim of this project is to compare the performance of Rsnet18 and EfficientNet-B0 while operating on the same datasets. Prior to this project, I did not have any exposure to training or testing deep learning models. In completing this project, I hoped to make a novel and timely contribution to analyzing classifications while learning how to perform deep learning research.

I was particularly interested in this project due to its timely relevance: Facial recognition has proven a cornerstone of new deep learning mass surveillance applications, and journalists, technology executives, and think tanks have recently argued that facial recognition should be regulated or controlled by the government [7].

II. LITERATURE REVIEW

At the time of my bachelor's in computer science in India, the pre-deep learning era was catching up. I was doing AI related database constraint analysis. My classmates were using Eigenfaces where Principal Component Analysis treated images as vectors to mathematically find the average face, but there were struggles. I was not in touch with AI because I was doing Data Engineering, but a deep learning breakthrough started to happen at that time. It came to the computer vision area too. Rsnet overcame the vanishing gradient challenge by introducing "skip connections" that allowed gradients to flow uninterrupted through layers. Then came the Neural Architecture Search. EfficientNet-B0 used it and emerged as a Baseline Network.

Explainable AI emerged in two distinct waves: first in the 1970s via transparent rule-based systems and later in the modern form around 2016. Grad-CAM emerged as the de-facto standard for explainable AI (XAI) for image classifications. It visually highlights exactly which parts of an image an AI model focuses on when making a decision.

III. DATA

Our facial recognition model is trained on the Real and Fake Face Detection from CIPLAB @ Yonsei University (Owner). License CC BY-NC-SA 4.0. It is available on Kaggle [6].



Fig. 1: Training sample from the Kaggle facial detection.

The dataset contains expert-generated high-quality photoshopped face images. The images are composite of different faces, separated by eyes, nose, mouth, or whole face. The objective of this project is to train a classifier for real and fake face images. In the case of generative models like Generative Adversarial Networks (GAN), it is very easy to generate fake face images. Then, a classifier can be trained using those images, and they do a great job discriminating between real and generated face images.

We can easily assume that the classifier learns some kind of pattern between images generated by GANs. However, those patterns can be futile in front of human experts, since exquisite counterfeits by experts are created in a completely different process. Thus, this dataset has expert level fake face photos.

We also used other Kaggle datasets. xhlulu/140k-real-and-fake-faces, mtesconi/twitter-deep-fake-text, manjilkarki/deepfake-and-real-images, sanikatiwarekar/deep-fake-detection-dfd-entire-original-dataset, saurabhbagchi/deepfake-image-detection, etc.

IV. FACIAL RECOGNITION MODELS

We began by developing a facial recognition model. In this section, we briefly describe how this model breaks down the facial recognition pipeline into three key steps: pre-processing, learning, and classification.

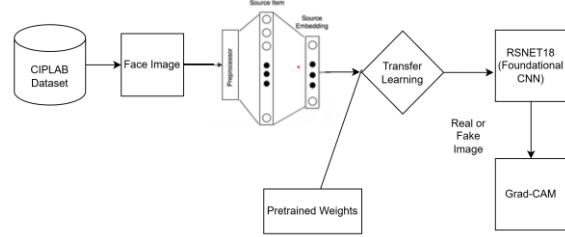


Fig. 2: Rsnnet-18 training flow [10].

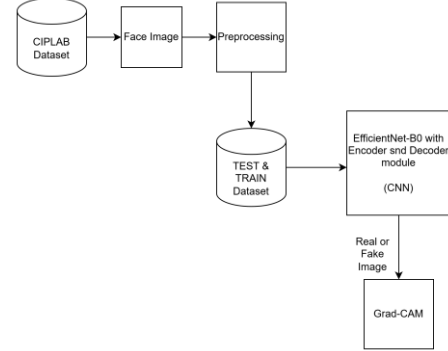


Fig. 3: EfficientNet-B0 training flow [8].

Let's first talk about Rsnnet18. In the preprocessing step, the model prepares images for facial recognition, which involves several steps. The first of which segmentation occurs by identifying the largest face present in any image, which is followed by the second step — cropping the image. The image is subsequently rotated and aligned so that each image has its respective facial features in the same pixel-region of the image.

The learning step of the model takes the pre-processed image and uses it to update weights in order to classify individual faces. This is done by generating 128-dimensional embeddings for each face. In order to create these embeddings, we use the Inception ResNet 18 (a convolutional neural network). This was desirable to us since the ability to use Batch-Norm on the auxiliary classifiers was favorable to our facial recognition task primarily for the purpose of regularization, as we wanted to avoid overfitting, especially towards the end of our training process [1]. Since we did not have the resources nor the time to fully train the model from scratch, we used a set of pre-

trained weights for our model.

Classification is subsequently performed by using 128-dimensional image embeddings as inputs to the Softmax classifier for its final prediction.

In the case of EfficientNet-B0 (a highly efficient convolutional neural network), we use the model to create the embeddings we already described. This architecture was desirable to us because its compound scaling method uniformly scales depth, width, and resolution, allowing for optimal feature extraction while remaining computationally lightweight. Since we did not have the resources or time to fully train the model from scratch, we too used a set of pre-trained weights for the training process. All other things are the same as for ResNet18.

We will explain here the last component, i.e. Grad-CAM algorithm.

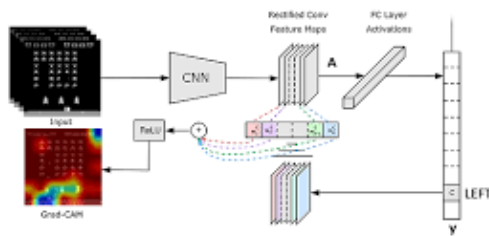


Fig. 4: Grad-CAM flow [9].

The input image is propagated forward through the CNN which produces feature maps. These feature maps are propagated through the fully connected layers of the A2C (Advantage. Actor, Critic) model. The gradient of the chosen action is backpropagated to the feature maps using a one-hot vector corresponding to the agent's chosen action, which constrains the backpropagated gradients to focus exclusively on the target class. These back-propagated gradients are then global-average pooled over the dimensions of the image to obtain importance weights for each feature. These weights are linearly combined with the feature map activations and rectified to obtain the Grad-

CAM heatmap.

EfficientNet-B0 generally yields better results with Grad-CAM. Because of its compound scaling strategy, it accurately zeroes in on specific features and provides highly focused heatmaps. ResNet18 is easier to implement but can produce overly diffuse or scattered heatmaps on complex images.

V. RESULTS

Looking at the "Training with Scoring and Validation" matrices, EfficientNet-B0 seems to be performing better. We look at the Train Acc, Val Acc, and Val Loss, If we averaged out the Fake and Real confidence during prediction, the EfficientNet-B0 scored a bit higher. It also seemed that Grad-CAM focus is much more elaborate for EfficientNet-B0. Therefore, in all cases EfficientNet-B0 wins, even though training time for it is a bit longer. But on t4 GPU the extra training time did not seem very significant.

```
Detecting Deepfakes with ResNet18 & CNN, then use ExplainableAI
Runtime: t4 GPU
Rsnet18 with Grad-CAM

Import Kaggle Data (Time): 10m Total Size of Data: (23.407345) GB.

Importing Libraries: 2s

Load data: 1m
Training images: 100000 | Validation images: 20000
Detected 2 classes: ['fake', 'real']

Model Architecture: 1s

Training with Scoring and Validation: 20m

FINISH Epoch 1 -> Train Acc: 0.7847 | Val Acc: 0.8125 | Val Loss: 0.4060
FINISH Epoch 2 -> Train Acc: 0.8183 | Val Acc: 0.8117 | Val Loss: 0.4080
FINISH Epoch 3 -> Train Acc: 0.8303 | Val Acc: 0.8501 | Val Loss: 0.3440

Plot Accuracy and Loss: 40s

Prediction: 2s (Both Original face and Grad-CAM focus of the face was printed)
Fake(99.89% confidence)
Real (85.42% confidence)

Grad-CAM: Less Focus.
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Fig 5: Rsnet-18 performance.

```
Detecting Deepfakes with EfficientNet-B0 & CNN, then use ExplainableAI
Runtime: t4 GPU
EfficientNet-B0 with Grad-CAM

Import Kaggle Data (Time): 10m Total Size of Data: (23.407345) GB.

Importing Libraries: 2s

Load data: 2m
Training images: 100000 | Validation images: 20000
Detected 2 classes: ['fake', 'real']

Model Architecture: 0s
Training with Scoring and Validation: 24m

FINISH Epoch 1 -> Train Acc: 0.8036 | Val Acc: 0.8522 | Val Loss: 0.3377
FINISH Epoch 2 -> Train Acc: 0.8402 | Val Acc: 0.8793 | Val Loss: 0.2903
FINISH Epoch 3 -> Train Acc: 0.8544 | Val Acc: 0.8887 | Val Loss: 0.2719

Plot Accuracy and Loss: 10s

Prediction: 1s (Both Original face and Grad-CAM focus of the face was printed)
Fake(98.91% confidence)
real (89.70% confidence)

Grad-CAM: More focus
```

Fig 6: EfficientNet-B0 performance.

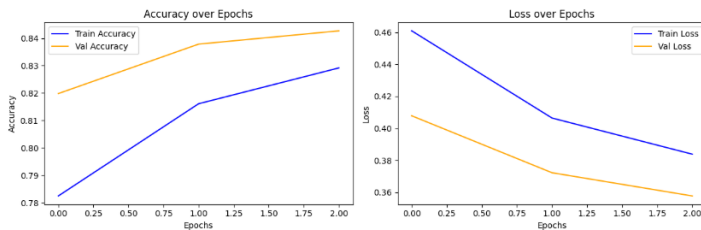


Fig 7: Rsnet-18 Accuracy and Loss.

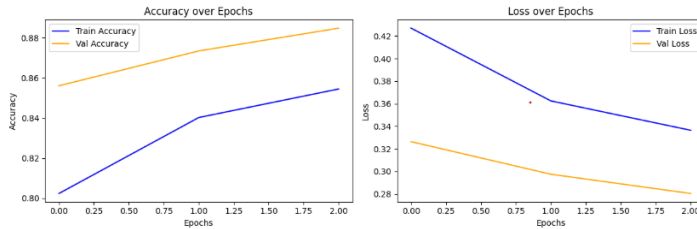


Fig 8: EfficientNet-B0 Accuracy and Loss.



Fig 9: Rsnet-18 with Grad-CAM, Top: Fake, Bottom: Real



Fig 10: EfficientNet-B0 with Grad-CAM, Top: Fake, Bottom: Real

VI. FUTURE WORK

Future work for ResNet18 and EfficientNet-B0 combined with Grad-CAM generally focuses on bridging the gap between high algorithmic accuracy and clinical or edge-deployment transparency. Because these lightweight convolutional neural networks (CNNs) are highly efficient, ongoing research continually refines how we interpret them [2].

1. **Moving Beyond Basic Grad-CAM:** Researchers are shifting to more advanced, higher-resolution techniques like Grad-CAM++, Eigen-GradCAM, and Score-CAM. These methods help remove background noise and provide tighter, more precise Bounding of features.
2. **Sanity Checks & Trust Alignment:** Future work continues to validate whether heatmaps truly reflect human reasoning. Studies are aimed at confirming that the regions highlighted by Grad-CAM actually align with the specific diagnostic biomarkers domain experts use [3].
3. **Video and 3D Volumetric Data:** Expanding 2D ResNet18 and EfficientNet-B0 architectures to process 3D medical scans (CT, MRI) or time-series video data (e.g., capsule endoscopy) while utilizing 3D Grad-CAM for spatial-temporal heatmaps [4].
4. **Hardware Optimization:** Because ResNet18 and EfficientNet-B0 are relatively lightweight, future goals include converting these models for use on low-resource hardware, mobile devices, and Point-of-Care (POC) medical tools.
5. **Adding Attention Blocks:** Modifying ResNet18 and EfficientNet-B0 to include specialized attention mechanisms (such as CBAM or Multi-Head Self-Attention). Future work studies how these attention layers enhance Grad-CAM's ability to pinpoint exact features [5].

VII. LINK TO COLAB NOTEBOOKS

Detecting Deepfakes with ResNet18 & CNN, then using ExplainableAI: Code is at <https://colab.research.google.com/drive/12w>

Ui5Gpci6sqWvitoEDish5tbCGalgAQ.

Detecting Deepfakes with EfficientNet-B0
& CNN, then using ExplainableAI: Code is at
https://colab.research.google.com/drive/1SGa4RFI8T_vDmX21_2BrX4J5-WbHqrj5

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- [8] EfficientNet architecture: https://www.researchgate.net/figure/Proposed-model-process-flow_fig2_357488896
- [9] Grad-CAM architecture: <https://dl.acm.org/doi/pdf/10.1145/3723498.3725723>
- [10] Rsnets-18 diagram was modified by me. I used Andrew Milich and Michael Karr's diagram as a baseline (Project name: Eluding Mass Surveillance: Adversarial Attacks on Facial Recognition Models).